PRODUCT DATASHEET · SKU PIA24027P

Coin-Sized 16-Channel Wireless EEG Sensor

Sixteen dry electrodes, a reference and a ground in one 26 mm round board: a cluster of spring-loaded pogo pins that press through the hair onto the scalp. Two TI ADS131M08 converters measure all 16 channels and a Seeed Studio XIAO ESP32-S3 streams them over Bluetooth LE straight to the browser. No gel, no cap, no ear clips.



16	18	Ø 26 mm	250 SPS	BLE 5
Channels + REF + GND	Dry pogo-pin electrodes	4-layer round board	24-bit, simultaneous	Straight to the browser

01 OVERVIEW

Key features

- **Electrodes built in**: 18 spring-loaded pogo pins (16 signal, REF, GND) in one ~20 × 20 mm cluster.
- **Seconds to set up**: no cap, no gel, no ear clips — strap it on and stream.
- **Two ADS131M08 converters**: 2 × 8 simultaneous 24-bit channels, DC-block filter, gain 4.
- **Wireless**: Bluetooth LE 5 (2M PHY) through a XIAO ESP32-S3; appears as Octopus-16-XXXX.
- **Browser-first software**: PiEEG Cloud, octoBridge, JavaScript and Python SDKs, Garmin octoWatch.
- **Flash from the browser**: one-click firmware install at octobridge.pieeg.com/flash.

Typical applications

- Quick alpha and blink demos
- Classrooms and the OctoBio Kit
- Neurofeedback and BCI games
- Local "brain stethoscope" over Pz, Oz, Cz
- Web and wearable app development
- Local 16-point spatial maps

02 TECHNICAL SPECIFICATIONS

SIGNAL ACQUISITION	
Converters	2 × Texas Instruments ADS131M08, 24-bit, 8-channel, simultaneous-sampling delta-sigma
Channels	16 (8 per converter), common reference and ground in the same cluster
Gain	4 on all channels (set in firmware; ADS131M08 supports 1 to 128)
DC block	ADS131M08 built-in DC-block filter, about 1 Hz
Reference voltage	1.2 V internal · 1 LSB ≈ 0.0358 µV at gain 4
Clock / sample rate	External 8.192 MHz oscillator, OSR 16384 → 250 SPS
ELECTRODE ARRAY	
Contacts	18 dry spring-loaded pogo pins, 4 × 11 mm: 16 signal, 1 reference (AINREF), 1 ground
Geometry	Hexagonal, pitch ≈ 5.38 mm, span ≈ 20.3 × 20.3 mm, all pins within 10.5 mm of centre
Reference / ground	Selectable by jumpers for single-board or stacked use
WIRELESS & CONTROLLER	

Controller	Seeed Studio XIAO ESP32-S3 with BLE antenna (plugs into two 7-pin sockets)
Radio	Bluetooth LE 5, 2M PHY · advertised name Octopus-16-XXXX
Data	51-byte notification per sample: 0xA0, counter, 16 × 3 bytes, 0xC0
POWER & SAFETY	
Supply	Small 5 V power bank via the XIAO USB-C, ≤ 2 A
Isolation	None — battery only while worn
Classification	Research, education and experimentation. Not a medical device
MECHANICAL	
PCB	Ø 26.0 mm round, 4 layers, 1.6 mm FR-4, black mask
Height	11 mm pogo pins below the board; XIAO stacks on top
Weight	60 g (shipping, Elecrow listing)
3D models	Holder on Thingiverse (thing 7386761); STEP/IGES models available
SOFTWARE	
Browser	PiEEG Cloud · octoBridge · JavaScript SDK (device: 'octopus-16') — Chrome or Edge
Desktop / wearable	Python SDK (bleak) on Windows, macOS, Linux · octoWatch for Garmin

03 BLOCK DIAGRAM

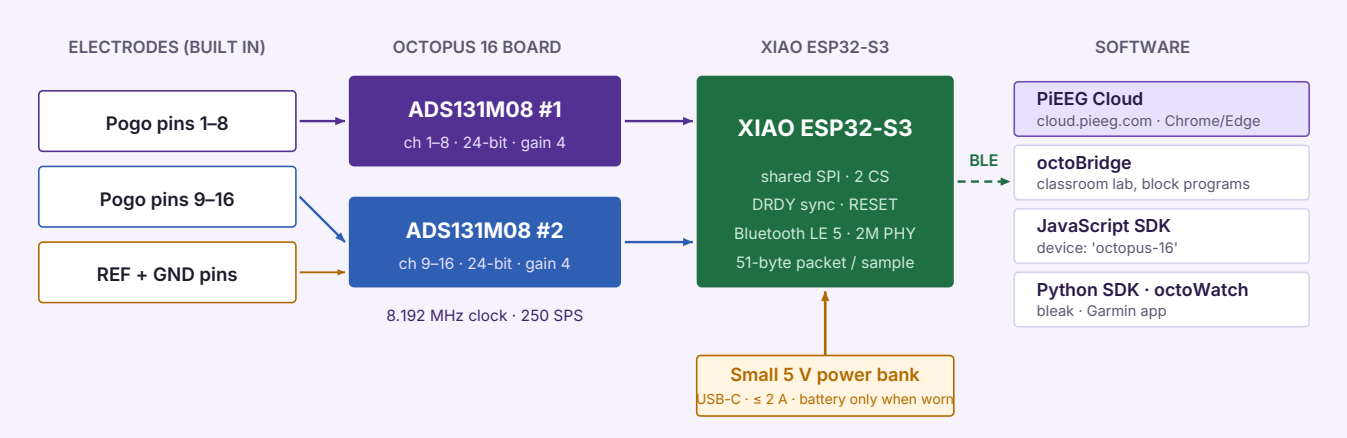


Figure 1 — Signal path from the pogo pins through the two converters and the XIAO to Bluetooth LE.

04 OCTOPUS 16 OR A CAP-BASED BOARD?

	PiEEG-16 / IronBCI	Octopus 16
Electrodes	Separate electrodes in a cap, ear clips	**Built in: 18 pogo pins**
Coverage	Spread over the whole head	**One local spot (~2 × 2 cm), e.g. over Pz**
Setup time	Minutes (cap, gel, clips)	**Seconds — strap it on**
Wireless	PiEEG-16: no · IronBCI: yes	**Yes, BLE 5**
Best for	Full-head research	**Quick tests, classrooms, neurofeedback**

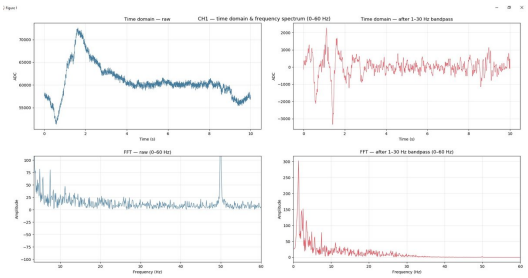


Figure 2 — Alpha with eyes closed, sensor over Pz.

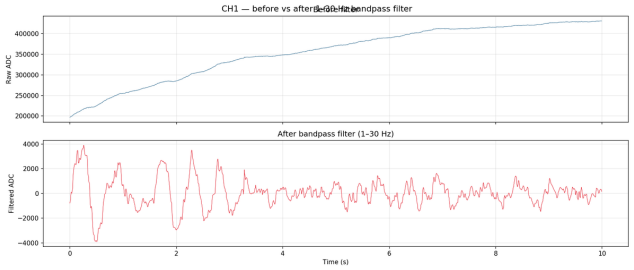


Figure 3 — Blink test over Fp, raw and filtered.

05 ORDERING INFORMATION

Product	Octopus 16 — 16-channel wireless EEG sensor
SKU	PIA24027P
Price	USD 250
Purchase	pieeg.com/store/octopus-16 · Elecrow (official manufacturing partner)
Kit	OctoBio Kit: Octopus 16 + Arduino classroom starter kit

Included	
1 × Octopus 16	18 pogo pins, 2 × ADS131M08
XIAO ESP32-S3 + antenna	Elecrow package; check your order

Not included	
Small 5 V power bank	≤ 2 A, battery only
USB-C data cable	For flashing
Strap or headband	Or 3D-printed holder

WARNING Power bank only while worn.

Octopus 16 has no isolation from its power supply. While it is on a head it runs from a small 5 V power bank only — never plugged into a computer or a wall charger. USB from a computer is for flashing only, with the board off the head. Octopus 16 is not a medical device.

06 RELATED DOCUMENTS

OC16-MD-002 Mechanical & Dimensional Drawing	**OC16-PCB-003** PCB & Manufacturing Specification	**OC16-PIN-004** Electrode Array & Pinout Guide
OC16-QSG-005 Quick Start & Safety Guide	**OC16-SW-006** Bluetooth Interface & Software Guide	